

Lecture: 9
Binary Trees

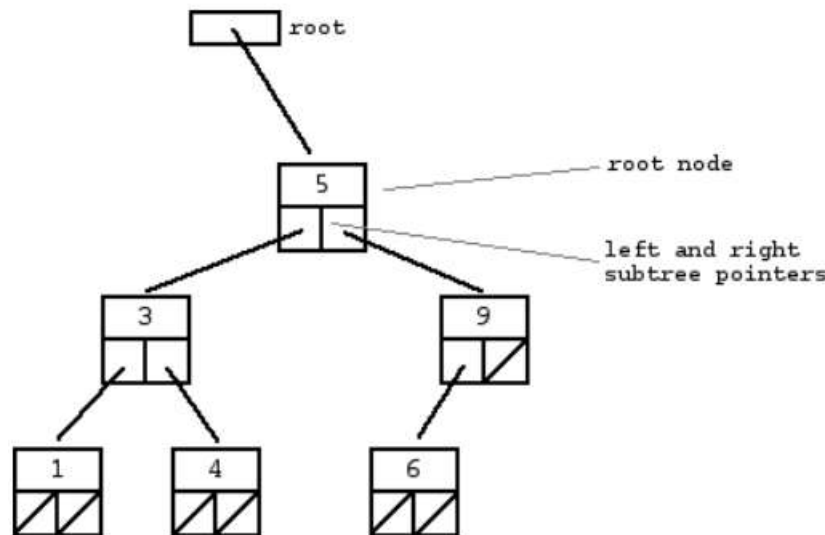
In a normal tree, every node can have any number of children. Binary tree is a special type of tree data structure in which every node can have a maximum of 2 children. One is known as left child and the other is known as right child. Thus, a tree in which every node can have a maximum of two children is called as Binary Tree.

- **In a binary tree, every node can have either 0 children or 1 child or 2 children but not more than 2 children.**
- A binary tree node contains following parts:
 - Data
 - Pointer to left child
 - Pointer to right child

A binary tree is made of nodes, where each node contains a “left” pointer, a “right” pointer and a data element. The “root” pointer points to the topmost node in the tree. The left and right pointers recursively point to smaller “subtrees” on either side.

A null pointer represents a binary tree with no elements—the empty tree.

“A binary tree is either empty (represented by a null pointer) or is made of a single node, where the left and right pointers (recursive definition ahead) each point to a binary tree.



There are different types of binary trees and they are:

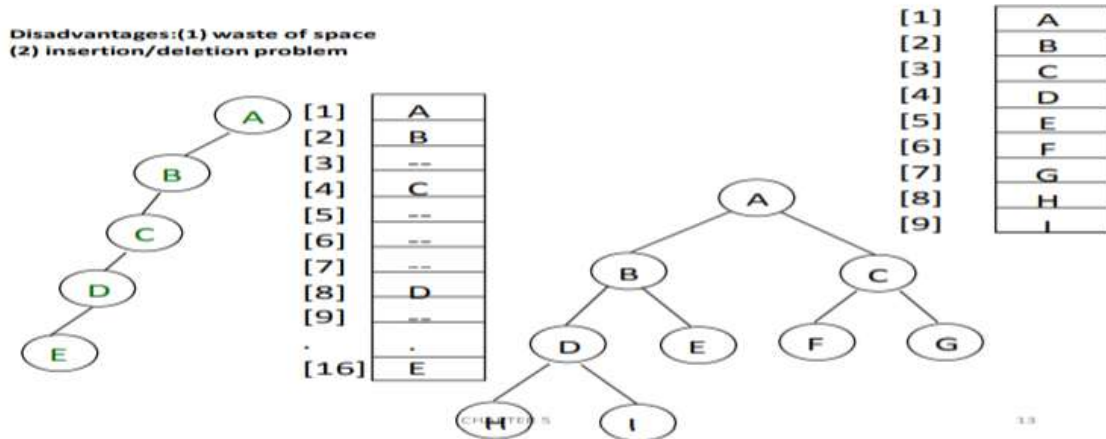
- **Strictly Binary Tree:** In a binary tree, every node can have a maximum of two children. But in strictly binary tree, every node should have exactly two children or none i.e. every internal node must have exactly two children. A strictly binary tree can be defined as follows:
 - A binary tree in which every node has either two or zero number of children is called strictly binary tree.
 - Strictly binary tree is also called as Full Binary Tree or Proper Binary Tree or 2-Tree.
- **Complete Binary Tree:** In a complete binary tree all the nodes must have exactly two children and at every level of complete binary tree there must be 2 level number of nodes. e.g. at level 2 there must be $2^2 = 4$ nodes and at level 3 there must be $2^3 = 8$ nodes.
 - A binary tree in which every internal node has exactly two children and all leaf nodes are at the same level is called Complete Binary Tree.

Binary Tree Representation:

- **Array Representation**
- **Linked List Representation**

Array Representation:

- In array representation of binary tree, we use a one dimensional array (1-D Array) to represent a binary tree.
- In general, array representation of a tree with depth d will require an array with approximately 2^{d+1} elements.
 - → this representation is usually inefficient unless the binary tree T is complete or nearly complete.
- This representation uses only a single linear array TREE as follows:
 - The root R of T is stored in TREE[1].
 - If a node N occupies TREE[K], then its left child is stored in TREE[$2*K$] and its right child is stored in TREE[$2*K+1$].
- Tree[1]= Null indicates that the tree is empty.



Linked Representation:

- In a linked list, every node consists of three fields.
 - First field, for storing left child address, second for storing actual data and third for storing right child address.
- In this linked list representation, a node has the following structure:

